DISTRIBUTION OF THE ESTIMATORS

Given $\underline{Y} = (Y_1, \dots, Y_n)$ and X $(n \times p)$

we have seen that the estimators are
$$\underline{\hat{B}} = (X^T X)^{-1} X^T \underline{Y}$$
$$\hat{\Sigma}^2 = \frac{1}{n} \underline{E}^T \underline{E} \qquad \text{with } \underline{E} = \underline{Y} - \underline{\hat{Y}} = \underline{Y} - X\underline{\hat{B}}$$
$$S^2 = \frac{1}{n-p} \underline{E}^T \underline{E}$$

we want to derive their exact distribution to perform inference.

• DISTRIBUTION OF $\underline{\hat{B}}$

estimator $\underline{\hat{B}} = (X^T X)^{-1} X^T \underline{Y}$ linear in $\underline{Y}$ (i.e. $\underline{\hat{B}} = A \cdot \underline{Y}$) with $\underline{Y} \sim N_n(X\underline{\beta}, \sigma^2 I_n)$

- LINEAR TRANSFORMATIONS of GAUSSIAN RANDOM VECTORS

$\underline{z} \sim N_d(\underline{\mu}, \Sigma)$, A $(k \times d)$ matrix, $\underline{b} \in \mathbb{R}^k$
$$\Rightarrow \quad T = \underbrace{\underset{k\times d}{A} \cdot \underset{d\times 1}{\underline{z}}}_{k\times 1} + \underset{k\times 1}{\underline{b}} \sim N_k(A\underline{\mu} + \underline{b}, \; A \Sigma A^T)$$

$$\mathbb{E}[\underline{T}] = \mathbb{E}[A\underline{z} + \underline{b}] = A\,\mathbb{E}[\underline{z}] + \underline{b} = A\underline{\mu} + \underline{b}$$
$$\text{var}(\underline{T}) = \text{var}(A\underline{z} + b) = \text{var}(A\underline{z}) = \mathbb{E}[(Az - \mathbb{E}[Az])(Az - \mathbb{E}[Az])^T]$$
$$= \mathbb{E}[(Az - A\mu)(Az - A\mu)^T] = \mathbb{E}[A(z-\mu)(z-\mu)^T A^T] = A \Sigma A^T$$

$$\Rightarrow \quad \underline{\hat{B}} = A\underline{Y} \quad \text{with } A = (X^T X)^{-1} X^T \quad p \times n \quad \text{rank} = p$$

$$\mathbb{E}[\underline{\hat{B}}] = A\,\mathbb{E}[\underline{Y}] = (X^T X)^{-1} X^T\, \mathbb{E}[\underline{Y}] = \cancel{(X^T X)^{-1}}\, \cancel{X^T X}\underline{\beta} = \underline{\beta}$$
$$\text{var}(\underline{\hat{B}}) = \text{var}(A\underline{Y}) = A\,\text{var}(\underline{Y})\,A^T = A(\sigma^2 I_n)A^T = \sigma^2 AA^T = \sigma^2 \cancel{(X^T X)^{-1}}\, \cancel{X^T X}\,((X^T X)^{-1})^T = \sigma^2 (X^T X)^{-1}$$

$$\Rightarrow \quad \underline{\hat{B}} \sim N_p(\underline{\beta}, \; \sigma^2 (X^T X)^{-1})$$

the marginal distribution of the j-th component is $\hat{B}_j \sim N_1(\beta_j ; \; \sigma^2 [(X^T X)^{-1}]_{(j,j)})$ $\quad j = 1, \dots, p$

the covariance between two components j and s $\quad \text{cov}(\hat{B}_j, \hat{B}_s) = \sigma^2 [(X^T X)^{-1}]_{(j,s)}$ $\quad j, s = 1, \dots, p$

Notice that the variance is $\sigma^2 (X^T X)^{-1}$: once again, we need $(\underline{x}_1, \dots, \underline{x}_p)$ to be linearly independent.

indeed $(X^T X)^{-1} = \frac{1}{\det(X^T X)} \cdot [\dots]$

If they are linearly dependent, $\det(X^T X) = 0$ and $(X^T X)$ is not invertible

Remark: if they are "almost collinear" $(\det(X^T X) \approx 0)$ the variance of the estimator explodes
$$\text{var}(\underline{\hat{B}}) = \sigma^2 \cdot \underbrace{\frac{1}{\det(X^T X)}}_{\to +\infty} \cdot [\dots]$$

• DISTRIBUTION OF THE RESIDUALS

$$\underline{e} = \underline{y} - \underline{\hat{y}} = \underline{y} - X\underline{\hat{\beta}} = \underline{y} - X \cdot (X^T X)^{-1} X^T \underline{y} = (I_n - \underbrace{X(X^T X)^{-1} X^T}_{\substack{\text{we called it } P:\\ P = X(X^T X)^{-1} X^T}})\,\underline{y} = (I_n - P)\,\underline{y}$$

$(n \times n)$ matrix

Let us study the corresponding random quantity $\underline{E}$
$$\underline{E} = (I_n - P)\underline{Y}$$
$$= (I_n - P)(X\underline{\beta} + \underline{\varepsilon})$$
$$= \underbrace{\cancel{(I_n - P)X\underline{\beta}}}_{=0} + (I_n - P)\underline{\varepsilon} = (I_n - P)\underline{\varepsilon} \qquad \underline{\varepsilon} \sim N(\underline{0}, \sigma^2 I_n)$$
$$\downarrow$$
$$(I_n - P)X\underline{B} = X\underline{B} - PX\underline{B} = X\underline{B} - \underbrace{X\cancel{(X^T X)^{-1}}\cancel{X^T X}}_{P}\underline{B} = X\underline{B} - X\underline{B} = 0$$

Hence, $\underline{E} = (I_n - P)\underline{\varepsilon}$ $\qquad \underline{\varepsilon} \sim N_n(\underline{0}, \sigma^2 I_n)$

linear combination of a Gaussian is Gaussian $\rightarrow$ $\underline{E} \sim N_n(\dots, \dots)$

$\mathbb{E}[\underline{E}] = (I_n - P)\underline{0} = \underline{0}$

$\text{var}(\underline{E}) = (I_n - P)\,\text{var}(\underline{\varepsilon})\,(I_n - P)^T = \sigma^2 (I_n - P)(I_n - P)^T = \sigma^2 (I_n - P)$
$$\Rightarrow \quad \underline{E} \sim N_n(\underline{0}, (I_n - P)\sigma^2)$$

(i.e. $\text{var}(E_i) = \sigma^2 (I_n - P)_{(ii)}$ $\rightarrow$ not homoscedastic)

• DISTRIBUTION OF $\hat{\Sigma}^2$

Preliminary result

If $\underline{z} \sim N_d(\underline{0}, \sigma^2 I_n)$; B a $(d \times d)$ matrix, symmetric, idempotent, with rank k $(1 \le k \le d)$
$$\Rightarrow \quad Q = \frac{1}{\sigma^2} \underline{z}^T B \underline{z} \qquad Q \sim \chi^2_k$$

$$\hat{\Sigma}^2 = \frac{1}{n} \underline{E}^T \underline{E}$$

$$\Rightarrow \quad n\hat{\Sigma}^2 = \underline{E}^T \underline{E} = ((I_n - P)\underline{\varepsilon})^T ((I_n - P)\underline{\varepsilon})$$
$$= \underline{\varepsilon}^T (I_n - P)\underline{\varepsilon}$$

We have seen that $(I_n - P)$ is symmetric and idempotent.

Moreover, the rank of $(I_n - P)$ is equal to $(n-p)$.

$\Rightarrow$ applying the result we get
$$\frac{1}{\sigma^2} \cdot \underline{\varepsilon}^T (I_n - P)\underline{\varepsilon} \sim \chi^2_{n-p} \qquad \Rightarrow \qquad \frac{\underline{E}^T \underline{E}}{\sigma^2} = \frac{n\hat{\Sigma}^2}{\sigma^2} \sim \chi^2_{n-p}$$

$n-p = \text{rank}(I_n - P)$
$= \#\text{units} - \#\text{covariates}$

Hence, we also find that $\mathbb{E}\left[\frac{n\hat{\Sigma}^2}{\sigma^2}\right] = n-p \Rightarrow \mathbb{E}[\hat{\Sigma}^2] = \frac{\sigma^2}{n}(n-p)$ biased

As usual, we obtain an unbiased estimator as
$$S^2 = \frac{\underline{E}^T \underline{E}}{n-p} = \frac{n\hat{\Sigma}^2}{n-p} \quad \text{with} \quad \frac{(n-p)S^2}{\sigma^2} \sim \chi^2_{n-p} \quad \Rightarrow \quad \mathbb{E}[\hat{S}^2] = \sigma^2$$

moreover, $\underline{\hat{B}} \perp\!\!\!\perp \hat{\Sigma}^2$ and $\underline{\hat{B}} \perp\!\!\!\perp S^2$